EE121 Discussion 7

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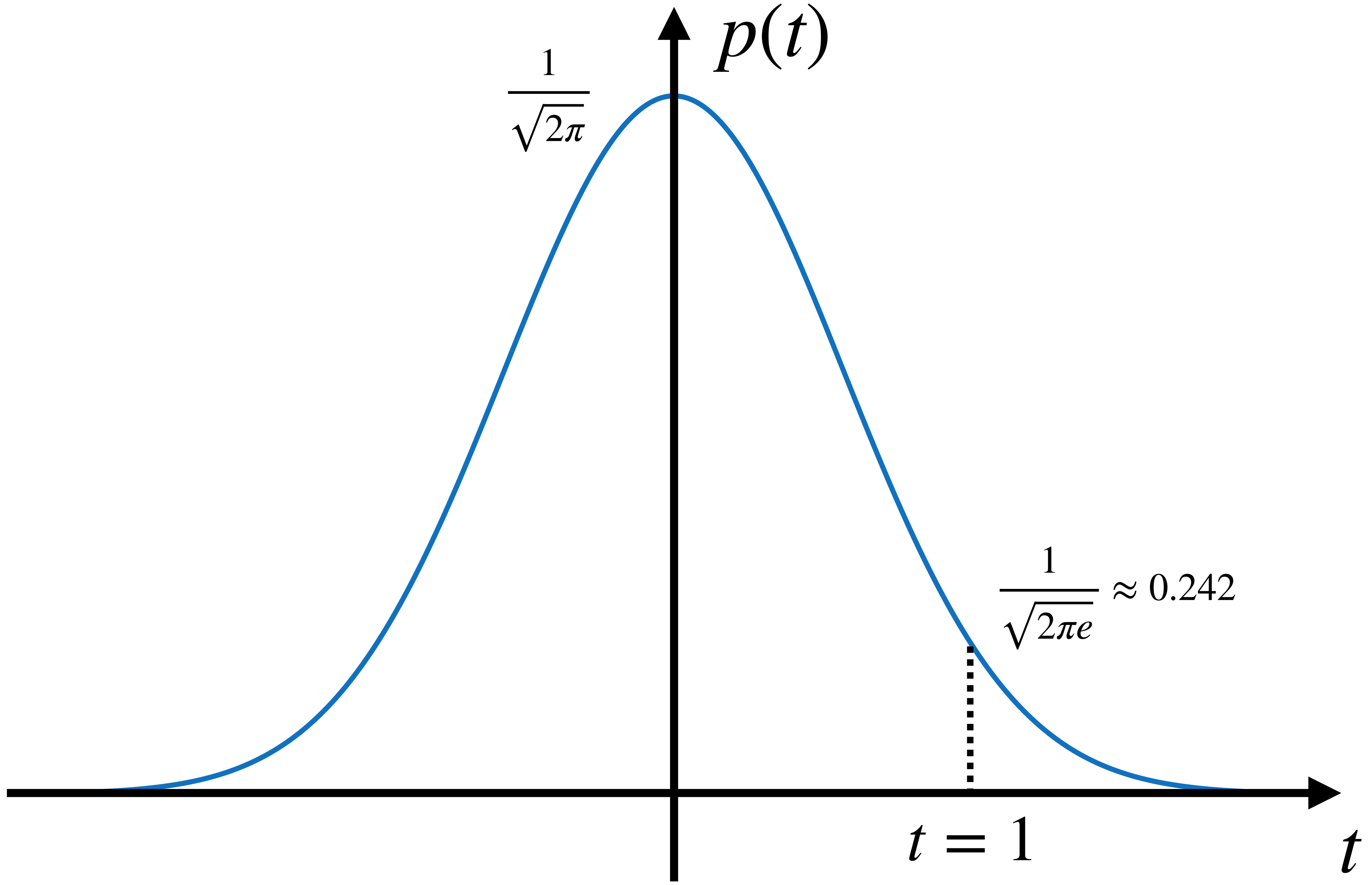
Agenda

- Q Function Practice
- Power Spectrum of Line Codes
- Raised Cosine Pulses
- Eye Diagrams
- QAM in Noise
- Repetition Codes
- Convolutional Codes

What's the Probability?

$$X \sim \mathcal{N}(0,1), \quad \begin{cases} P(X > x) = Q(x) = \int_x^{\infty} p(t)\, dt \\ P(X \leq x) = \Phi(x) \int_{-\infty}^{x} p(t)\, dt \end{cases}$$



What's $p(t)$?

$$p(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

How are $Q(x)$ and $\Phi(x)$ related?

$$\Phi(x) = 1 - Q(x)$$
$$\Phi(-x) = Q(x)$$
$$Q(-x) = 1 - Q(x)$$

I'm given $Y \sim \mathcal{N}(\mu, \sigma^2)$, can I use $\Phi(x)$ and $Q(x)$?

Yes, but you need to apply a transform to Y.

$$Y \sim \mathcal{N}(\mu, \sigma^2) \to (Y - \mu)/\sigma \sim \mathcal{N}(0,1)$$
$$P(Y > y) = P((Y-\mu)/\sigma > (y-\mu)/\sigma)$$
$$= P(\mathcal{N}(0,1) > (y-\mu)/\sigma)) = Q((y-\mu)/\sigma)$$

What's the Probability?

$$Y \sim \mathcal{N}(8,4),\ P(Y > 6) =$$

$$Y \sim \mathcal{N}(8,4) \rightarrow (Y-8)/2 \sim \mathcal{N}(0,1)$$

$$\begin{aligned} P(Y > 6) &= P((Y-8)/2 > (6-8)/2) \\ &= P(\mathcal{N}(0,1) > -1) = Q(-1) \end{aligned}$$

$$Y \sim \mathcal{N}(-3,1),\ P(Y \leq 0) =$$

$$Y \sim \mathcal{N}(-3,1) \rightarrow (Y+3) \sim \mathcal{N}(0,1)$$

$$\begin{aligned} P(Y \leq y) &= P((Y+3) \leq (y+3)) \\ &= P(\mathcal{N}(0,1) \leq 3) = Q(-3) \end{aligned}$$

$$Y \sim \mathcal{N}(-a, N_0),\ P(Y > -a/2) =$$

$$Y \sim \mathcal{N}(-a, N_0) \rightarrow (Y+a)/\sqrt{N_0} \sim \mathcal{N}(0,1)$$

$$\begin{aligned} P(Y > y) &= P((Y+a)/\sqrt{N_0} > (y+a)/\sqrt{N_0}) \\ &= P(\mathcal{N}(0,1) > a/(2\sqrt{N_0})) = Q(a/(2\sqrt{N_0})) \end{aligned}$$

Power Spectrum of Line Codes

$$S_y(f) = |P(f)|^2 \left(\frac{1}{T_b} \sum_n R_x[n] e^{-j2\pi f n T_b} \right)$$

$R_x[n]$

Polar Signaling $\quad R_x[n] = \delta[n]$

On-Off Keying (OOK) $\quad R_x[n] = \frac{1}{4}(1 + \delta[n])$

Split-phase, or Manchester coding $\quad R_x[n] = \delta[n]$

Bipolar signaling $\quad R_x[n] = \frac{1}{4}(2\delta[n] + \delta[n-1])$

*Equal Probabilities

Power Spectrum of Line Codes

$$S_y(f) = |P(f)|^2 \left(\frac{1}{T_b} \sum_n R_x[n] e^{-j2\pi f n T_b} \right)$$

$$P(f)$$

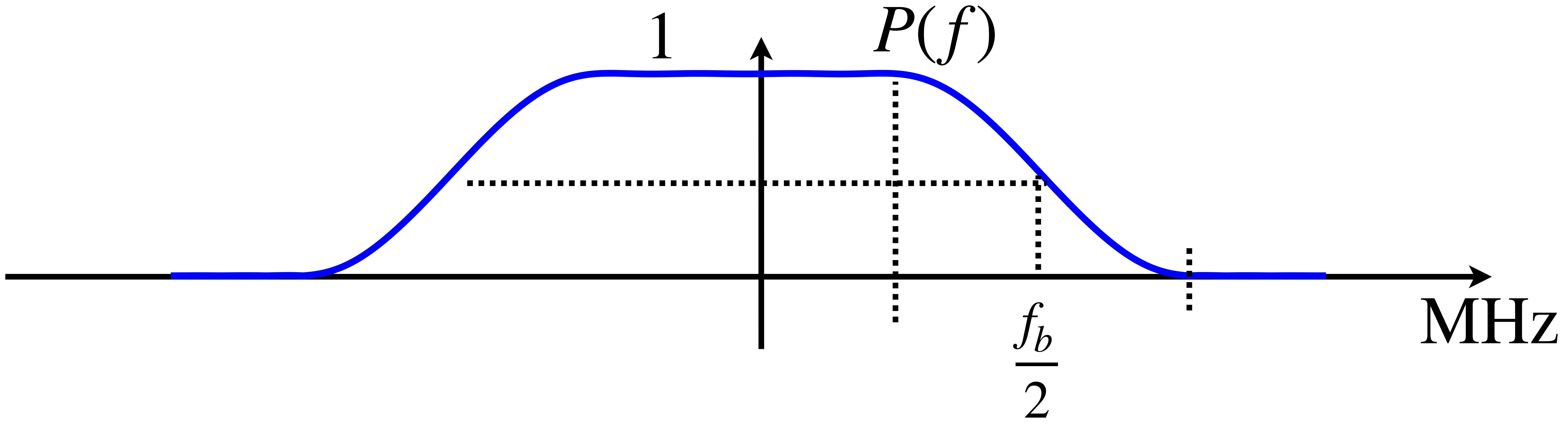
Raised Cosine Pulse

$$P(f) = \begin{cases} T_b, & |f| \le \frac{1-\beta}{2T_b} \\ \frac{T_b}{2}\left[1 + \cos\left((|f| - \frac{1-\beta}{2T_b})\frac{\pi T}{\beta} \right)\right], & \frac{1-\beta}{2T_b} \le |f| \le \frac{1+\beta}{2T_b} \\ 0, & |f| > \frac{1+\beta}{2T_b} \end{cases}$$

Sinc Pulse

$$P(f) = T_b \sqcap \left(fT_b\right)$$

Raised Cosine Pulse Practice



What is the BW using a RC pulse operating at 500Mbps using 64QAM and $\beta = 0.5$?

$$\tilde{f}_b = \frac{\text{Bit Rate}}{\log_2(M)} = \frac{600\ \text{Mbps}}{\log_2(64)} = 100\ \text{Mbaud} \qquad B = (1+\beta)B_{\text{min}} = 1.5 \times 100\ \text{MHz} = 150\ \text{MHz}$$

Raised Cosine Pulse Practice

Does this pulse has zero ISI at bitrate of 30Mbps using 8PSK?

$$p(t) = \text{sinc}\left(\frac{t}{T_b}\right)\frac{\cos(\pi\beta t/T_b)}{1-\left(\frac{2\beta t}{T}\right)^2}$$

Zero-crossing every:

$$\Rightarrow kT_b \quad \Rightarrow f_b/k \quad \Rightarrow k\frac{T_b}{\beta} \quad \Rightarrow \beta f_b/k$$

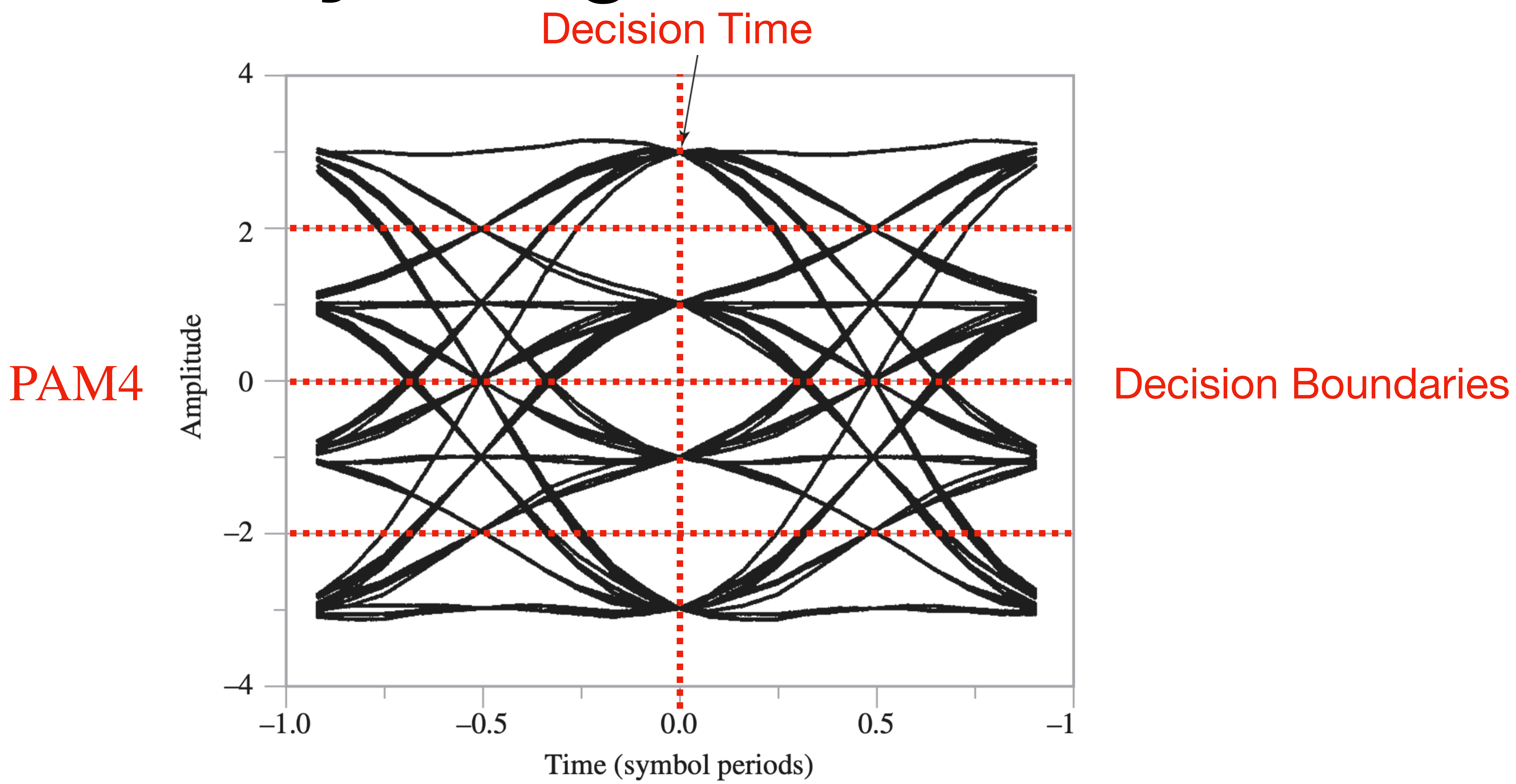
$$\tilde{f}_b = \frac{\text{Bit Rate}}{\log_2(M)} = \frac{30 \text{ Mbps}}{\log_2(8)} = 10 \text{ MHz}$$

$$B_{\min} = 10 \text{ MHz} \quad f_b = 18 \text{ MHz}$$

$$\Rightarrow 18 \text{ MHz}/k \qquad \Rightarrow 18 \text{ MHz}/(9k)$$

No!

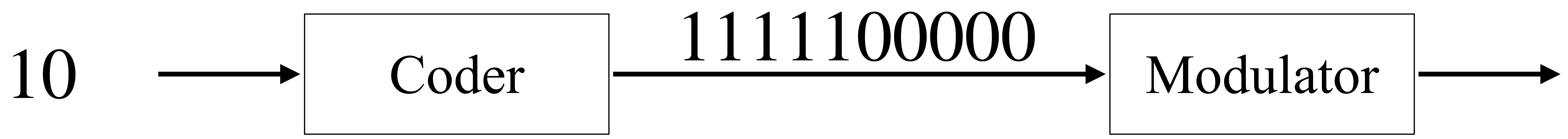
Decipher this Eye Diagram



What's the modulation scheme? When should I decide which symbol was sent?

Decision boundaries when each bit is equally likely?

Repetition Codes



What is the code rate?

$$r = \frac{1}{5}$$

Given that the modulation is PAM2 with $\mathrm{SNR} = 4$, what's the probability of error using a majority vote?

$$P_e(r = 1) = Q(2) = 0.0228 \triangleq p$$

$$P_e = \sum_{m=\lceil n/2 \rceil}^{n} \binom{n}{m} p^m (1-p)^{n-m}$$

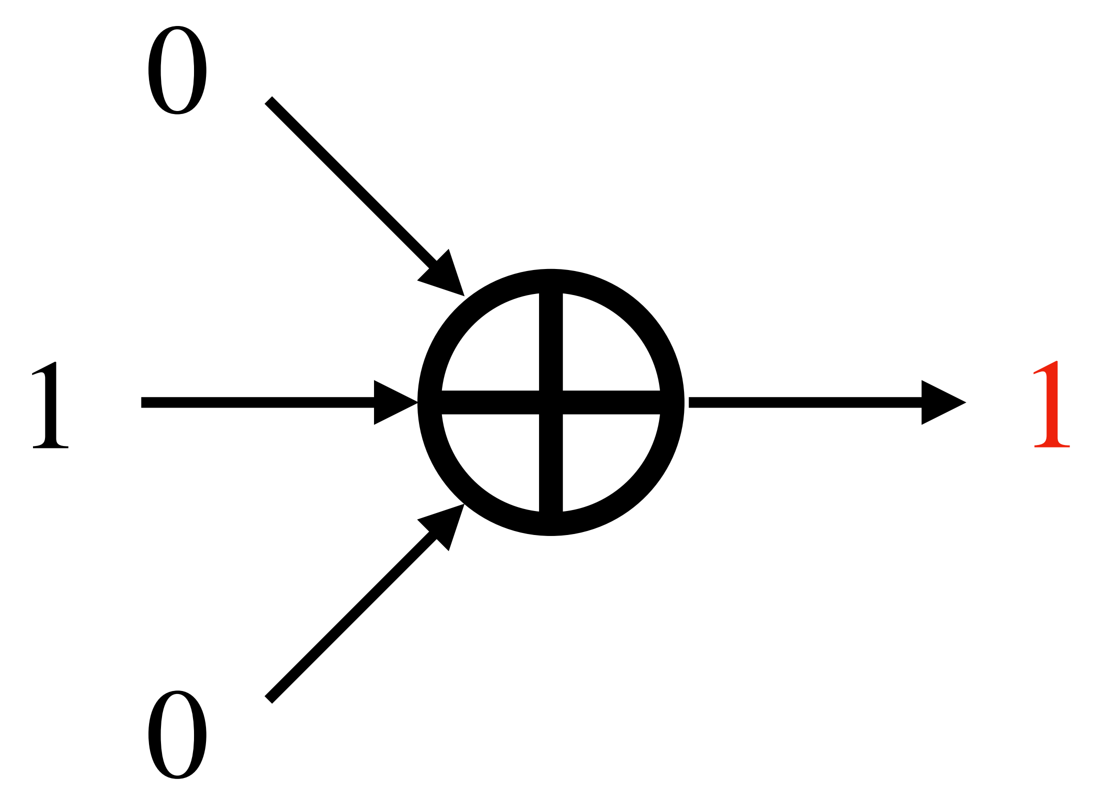
This is the case for odd n

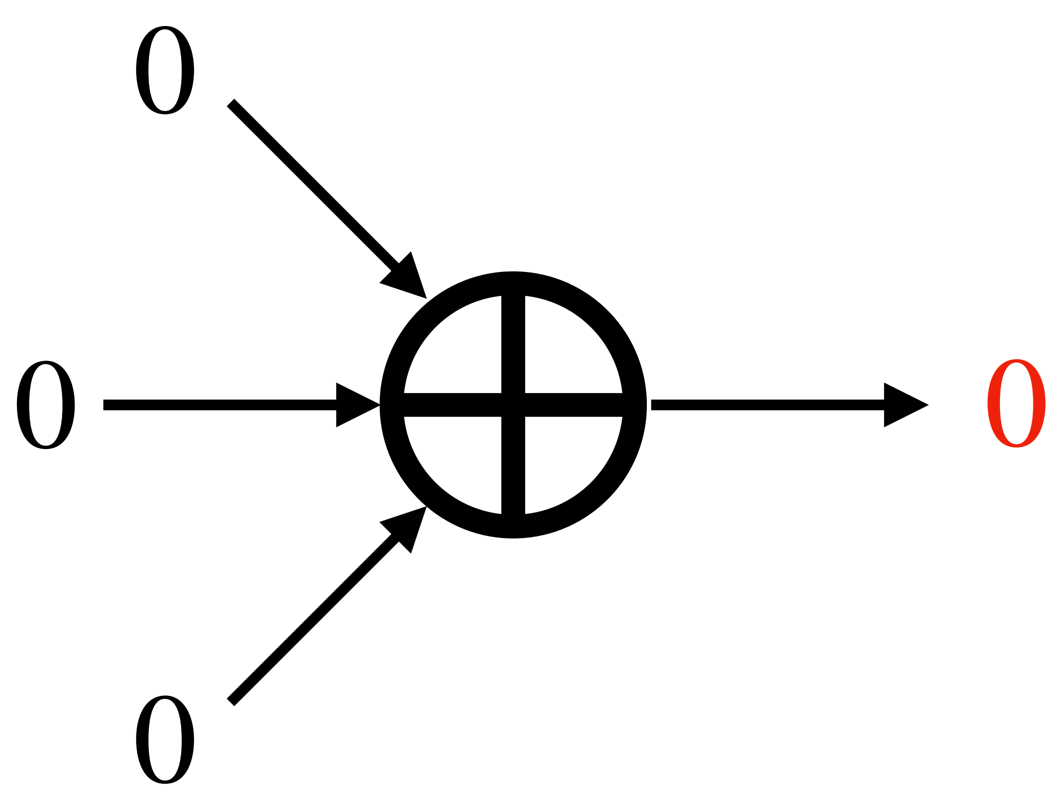
XOR Practice

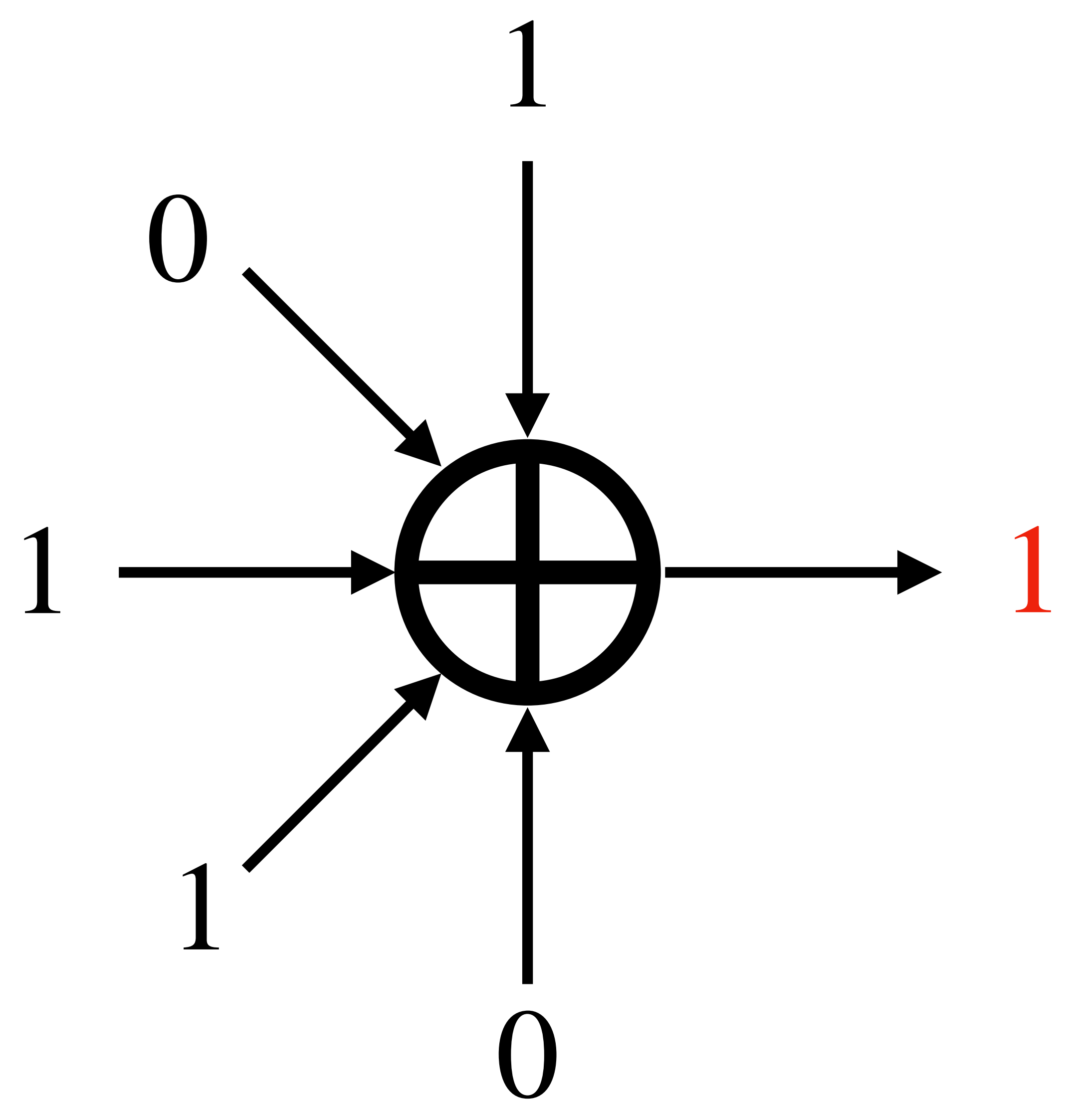
A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

A	B	C	$A \oplus B \oplus C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

ODD number of 1's







Convolutional Codes

Consider a code with generate $G(D)$ given as

$$G(D) = \begin{bmatrix} D & D & 1+D \\ D & 1 & 1 \end{bmatrix}$$

What's is the code rate?

Since $G(D)$ is a 2×3 matrix, the rate is $r = \dfrac{2}{3}$

Draw the corresponding encoder.

Convolutional Codes

Draw the corresponding encoder.

$$G(D) = \begin{bmatrix} D & D & 1+D \\ D & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}^T = [u_1 \quad u_2] \begin{bmatrix} D & D & 1+D \\ D & 1 & 1 \end{bmatrix}$$

$$y_1 = Du_1 + Du_2 = u_1[k-1] + u_2[k-1]$$

$$y_2 = Du_1 + u_2 = u_1[k-1] + u_2[k]$$

$$y_3 = (1+D)u_1 + u_2 = u_1[k] + u_1[k-1] + u_2[k]$$

Convolutional Codes

Draw the corresponding encoder.